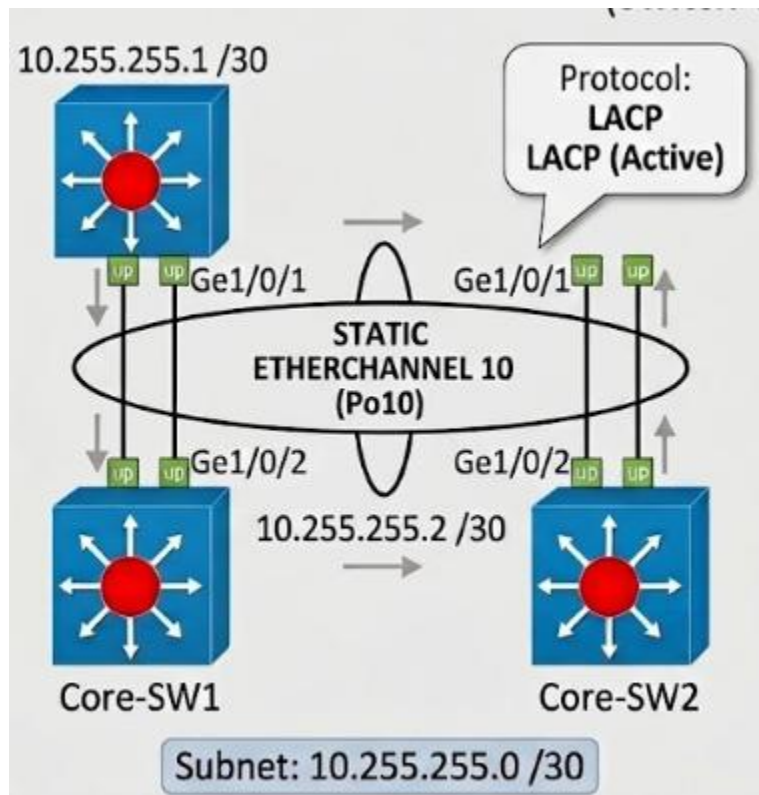


## Layer 3 LACP port-channel LAB



### Lab Topology

- **Device 1:** Core-SW1
- **Device 2:** Core-SW2
- **Physical Links:** GigabitEthernet 1/0/1 and GigabitEthernet 1/0/2 on both sides.
- **Port-Channel Number:** 10
- **IP Subnet:** 10.255.255.0/30

## Configure Core-SW1

On a Cisco multilayer switch, you must first enable IPv4 routing globally. Then, convert the physical ports to Layer 3 before bundling them to avoid mismatched port states.

```
SW1> enable
```

```
SW1# configure terminal
```

```
Sw1#hostname Core-SW1
```

Enable global IP routing (required on multilayer switches)

```
Core-SW1(config)# ip routing
```

### **New command**

**Default the interfaces to ensure a clean slate**

```
Core-SW1(config)# default interface range GigabitEthernet 1/0/1 - 2
```

```
Core-SW1(config)# interface range GigabitEthernet 1/0/1 - 2
```

```
Core-SW1(config-if-range)# no switchport
```

```
Core-SW1(config-if-range)# channel-protocol lacp
```

```
Core-SW1(config-if-range)# channel-group 10 mode active
```

```
Core-SW1(config-if-range)# no shutdown
```

```
Core-SW1(config-if-range)# exit
```

Configure the logical Port-Channel interface

```
Core-SW1(config)# interface Port-channel 10
```

```
Core-SW1(config-if)# no switchport
```

```
Core-SW1(config-if)# ip address 10.255.255.1 255.255.255.252
```

```
Core-SW1(config-if)# no shutdown
```

Core-SW1(config-if)# exit

## Configure Core-SW2

SW2> enable

SW2# configure terminal

SW2#hostname Core-SW2

Enable global IP routing

Core-SW2(config)# ip routing

### **New command**

#### **Default the interfaces**

**Core-SW2(config)# default interface range GigabitEthernet 1/0/1 - 2**

! Configure the physical interfaces

Core-SW2(config)# interface range GigabitEthernet 1/0/1 - 2

Core-SW2(config-if-range)# no switchport

Core-SW2(config-if-range)# channel-protocol lacp

Core-SW2(config-if-range)# channel-group 10 mode active

Core-SW2(config-if-range)# no shutdown

Core-SW2(config-if-range)# exit

! Configure the logical Port-Channel interface

```
Core-SW2(config)# interface Port-channel 10
```

```
Core-SW2(config-if)# no switchport
```

```
Core-SW2(config-if)# ip address 10.255.255.2 255.255.255.252
```

```
Core-SW2(config-if)# no shutdown
```

```
Core-SW2(config-if)# exit
```

## Verification Commands

```
show etherchannel summary
```

```
show ip interface brief | include Port-channel10
```